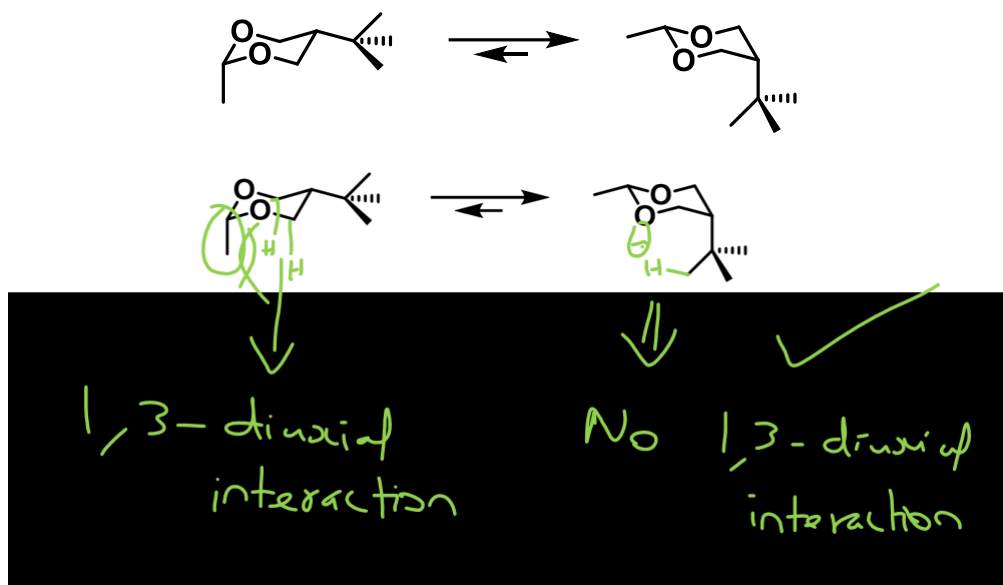


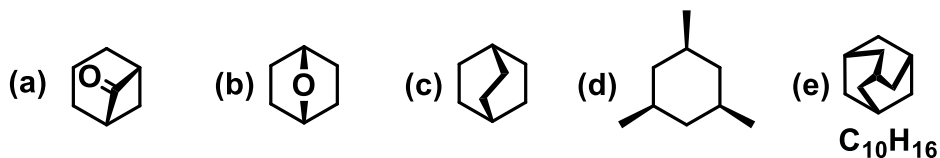
CHM102 Basic Organic Chemistry

Conformations, Aromaticity & Acidity/basicity Concepts

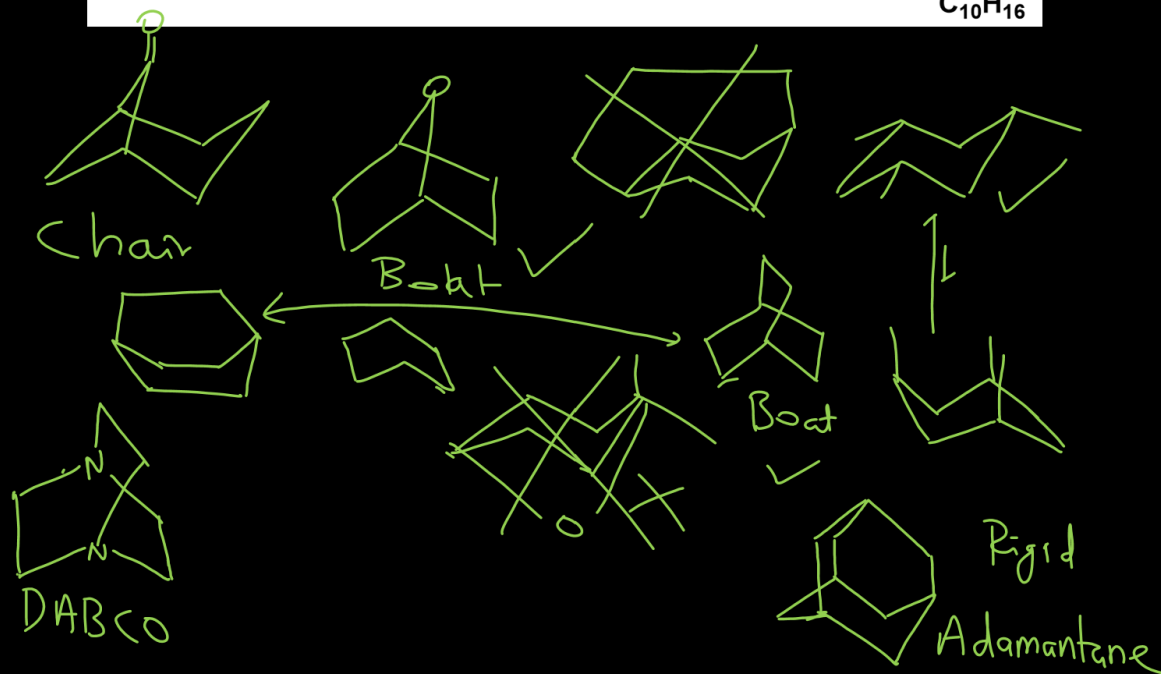
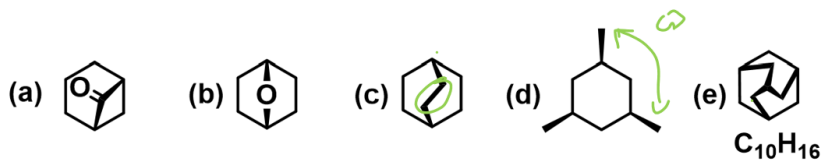
Q1. Cyclohexanes with bulky substituents in the equatorial position rather than the axial position are more stable. However, the following equilibrium shows the preference for bulky substituent in the axial position. What could be the reason?



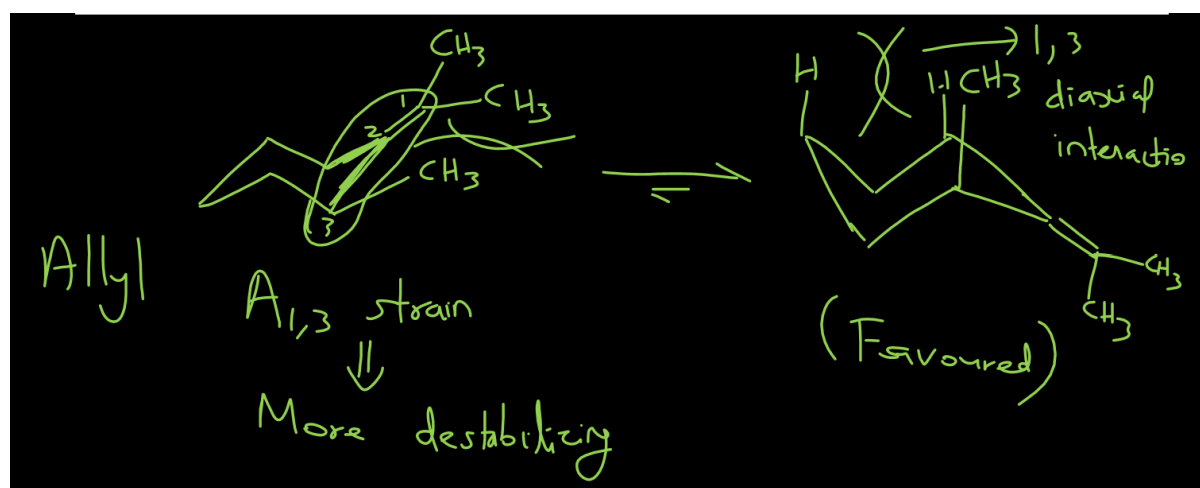
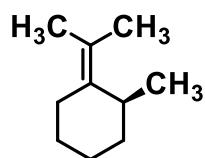
Q2. Draw the proper conformer of the following compounds.



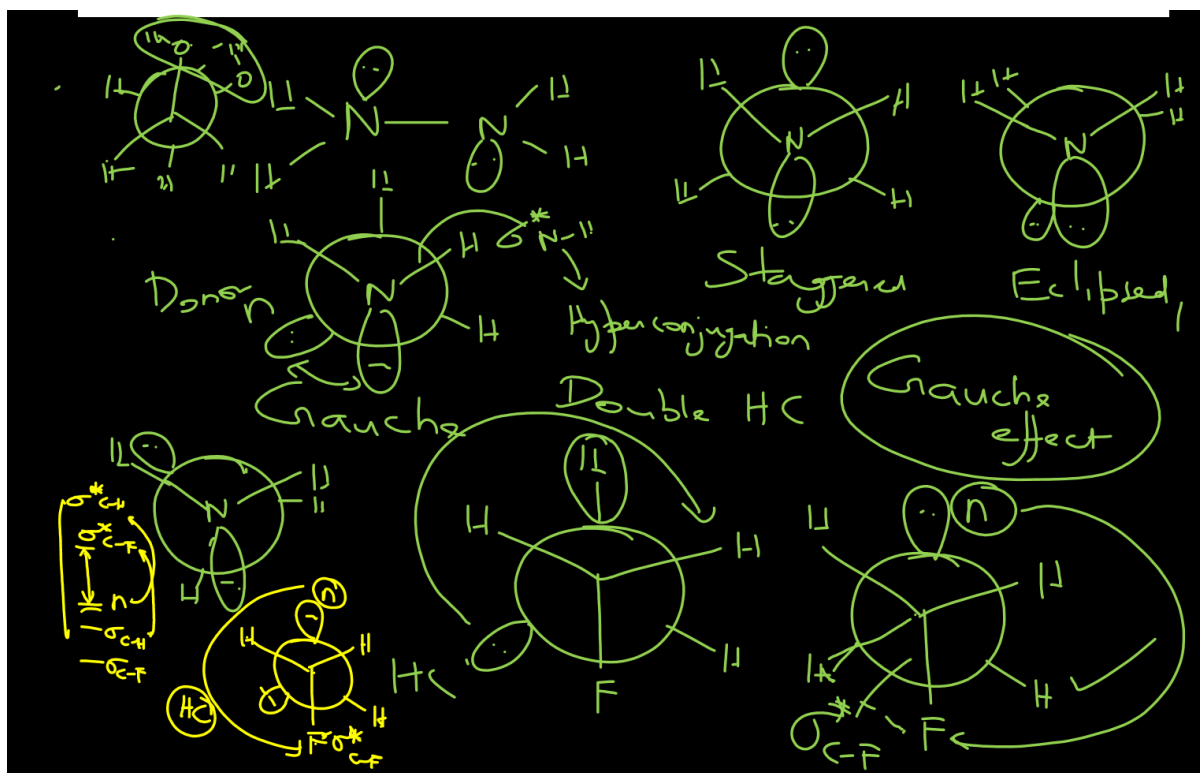
Q2. Draw the proper conformer of the following compounds.



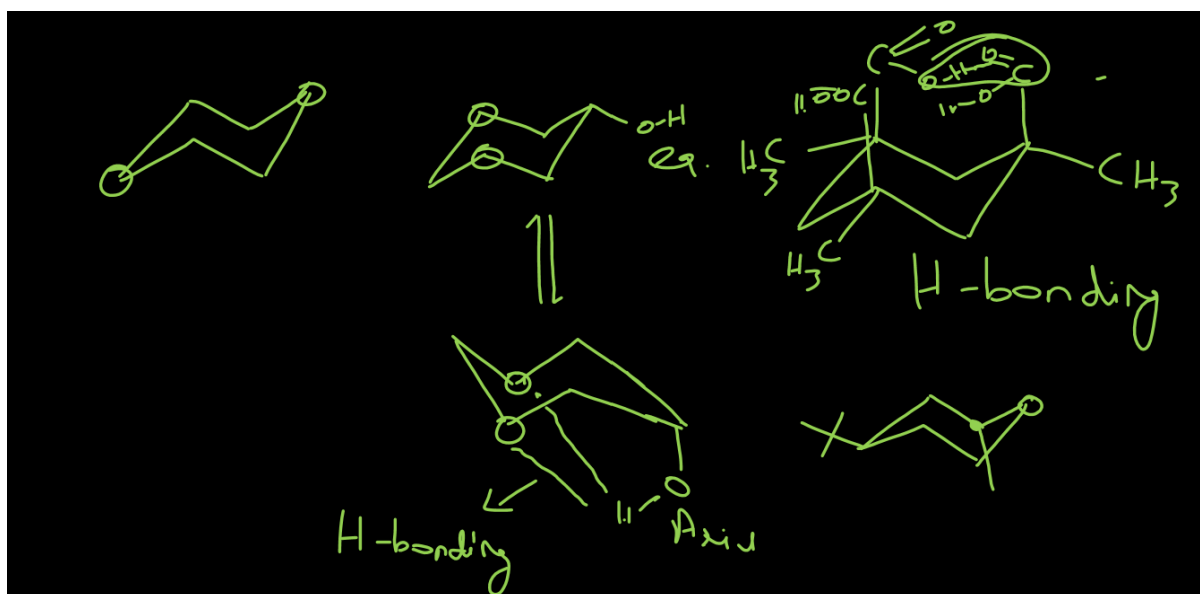
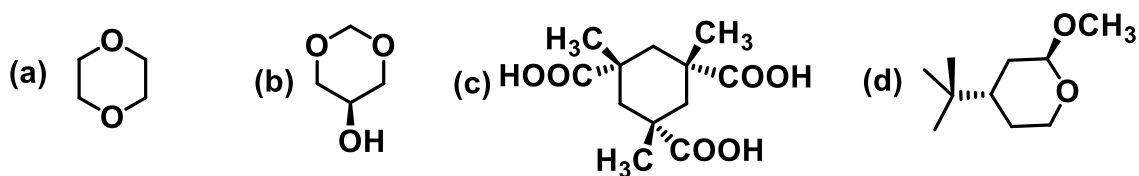
Q3. Draw the possible chair conformers of the following compounds. Which one of the chair forms is stable? Indicate the reason.



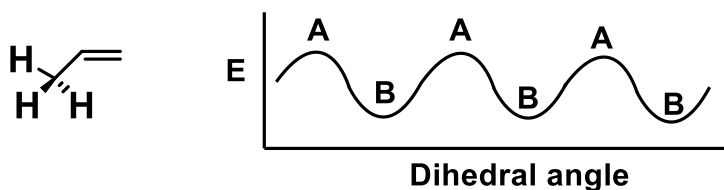
Q4. Indicate the possible conformers of hydrazine (NH_2NH_2), (fluoromethyl)amine (FCH_2NH_2) and fluoromethanol (FCH_2OH) using Newman projection and indicate the stability order with reasons.



Q5. Draw the most stable conformations for the following compounds:

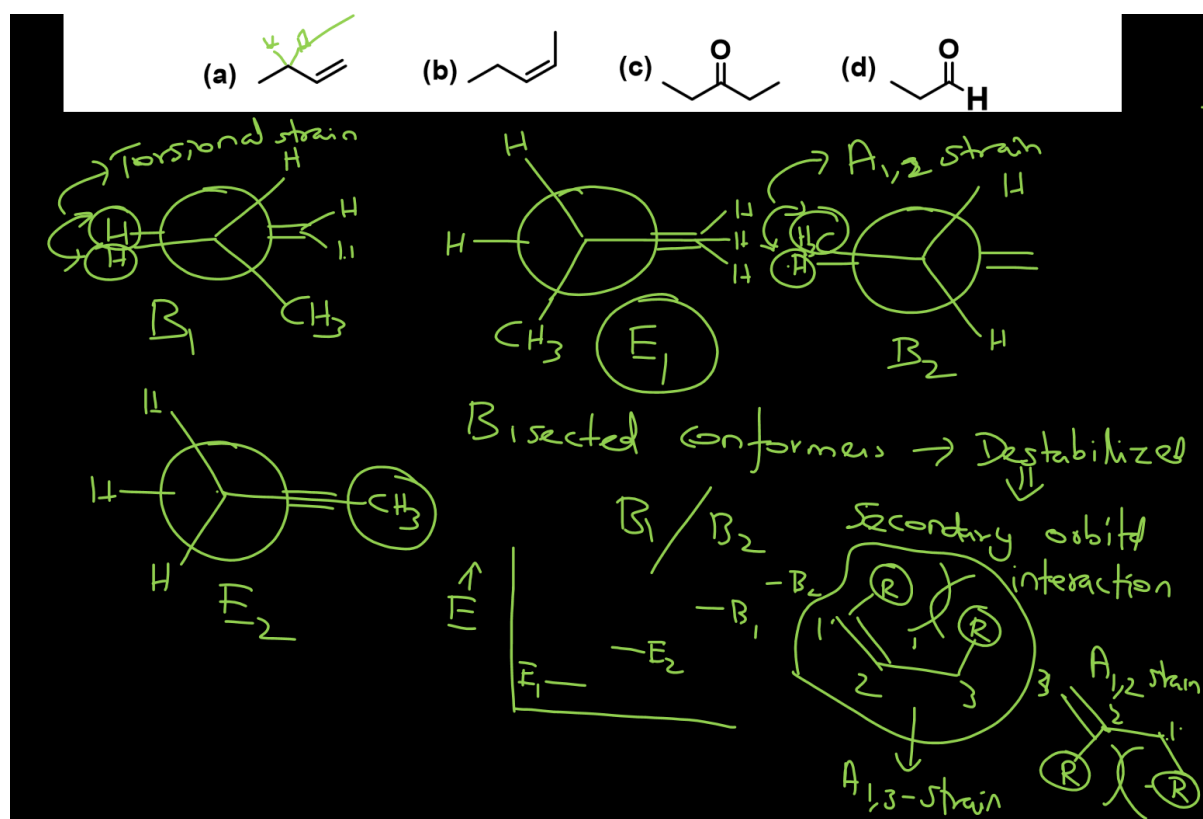
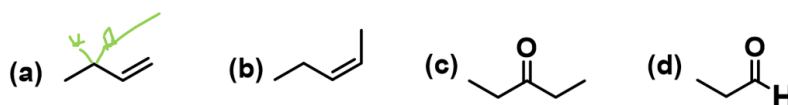
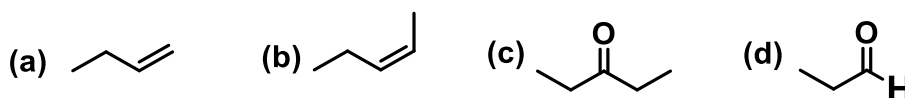


Q6. If the conformational changes can be represented for propene as follows, identify the conformations A and B and indicate the reason.

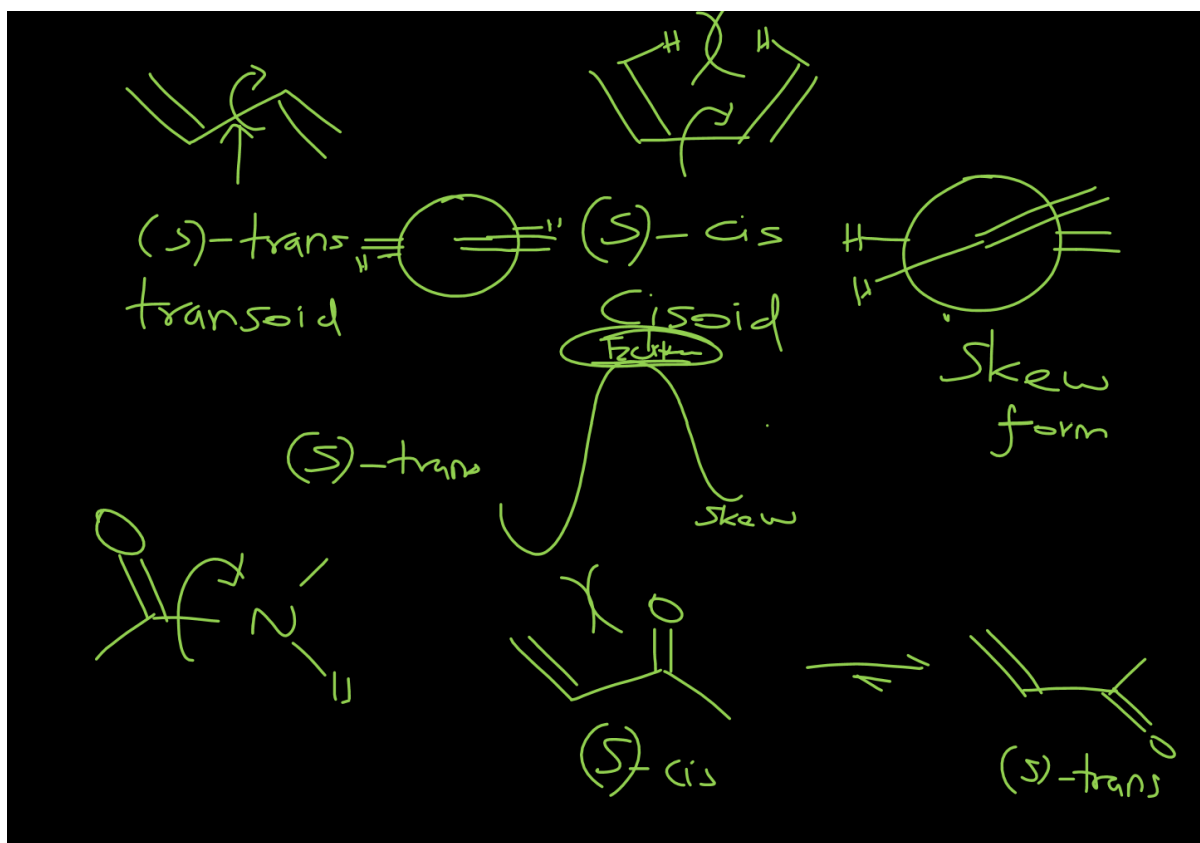


Solution: Refer to the class notes

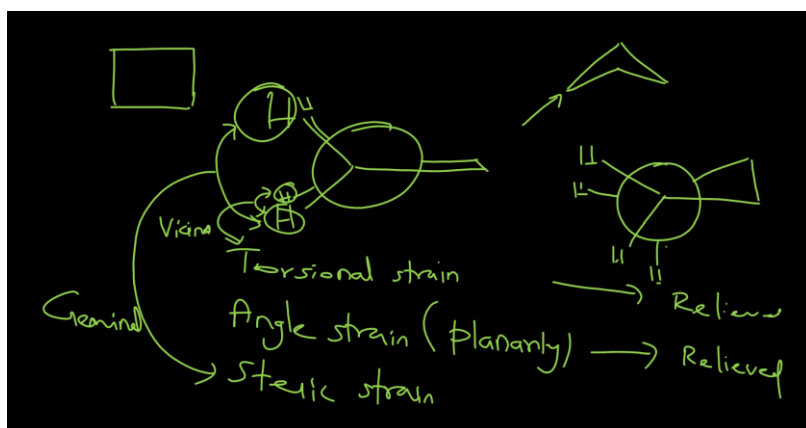
Q7. Identify the major conformers and indicate the most preferred conformations of the following compounds with reason:



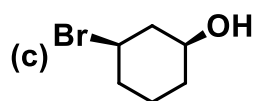
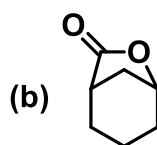
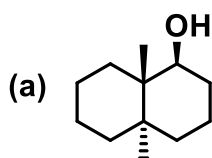
Q8. Identify the conformational preferences in 1,3-butadiene and 3-buten-2-one.

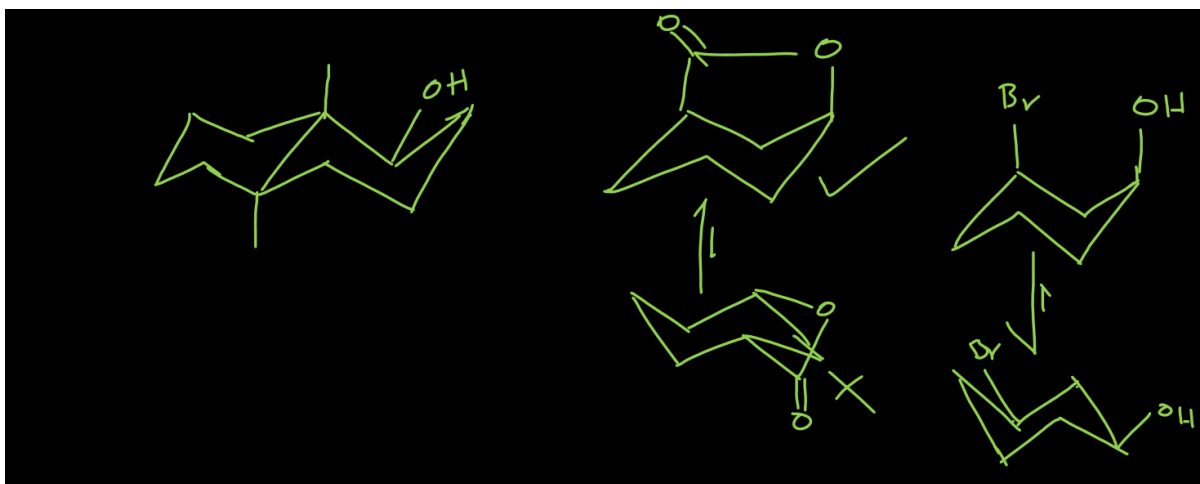


Q9. Draw the Newman projections of planar and puckered cyclobutane and compare the possible strains in each one of them.

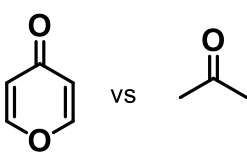
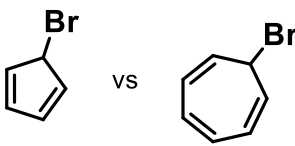
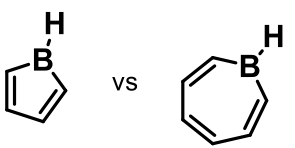
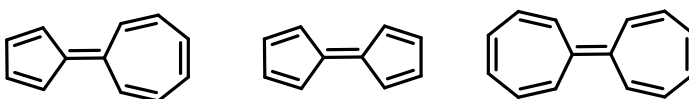


Q10. Draw the conformations of the following compounds.





Q11. Identify the molecule among each pair exhibiting the dominant property indicated for each case and indicate the reason.

- (a) Basicity 
- (b) Ionization of the C-Br bond 
- (c) Lewis acidity 
- (d) Dipole moment 

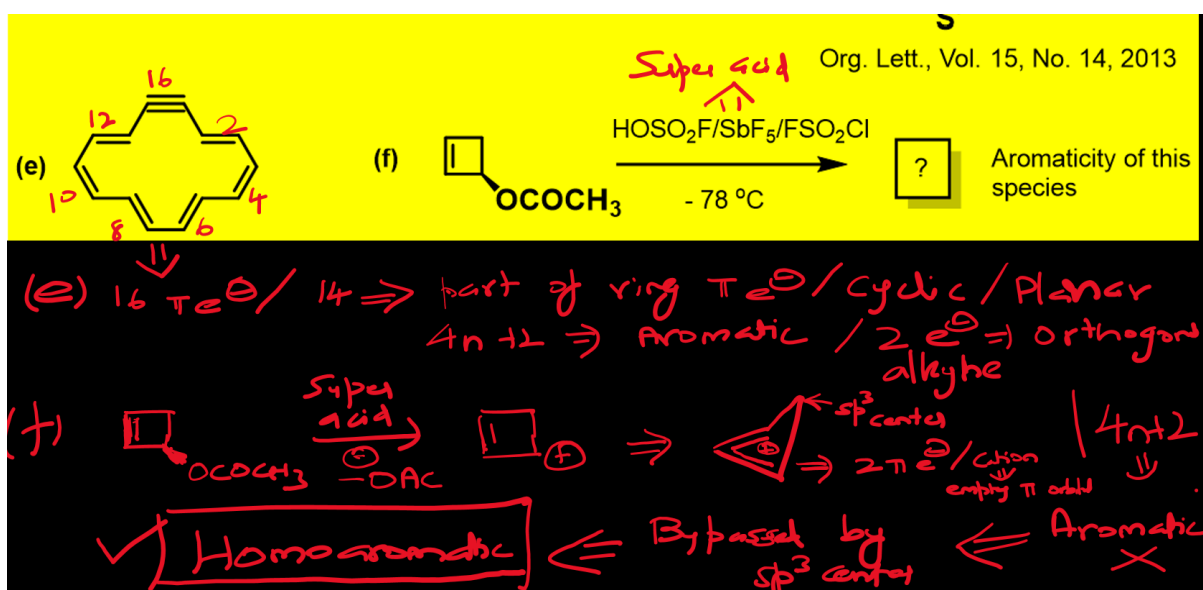
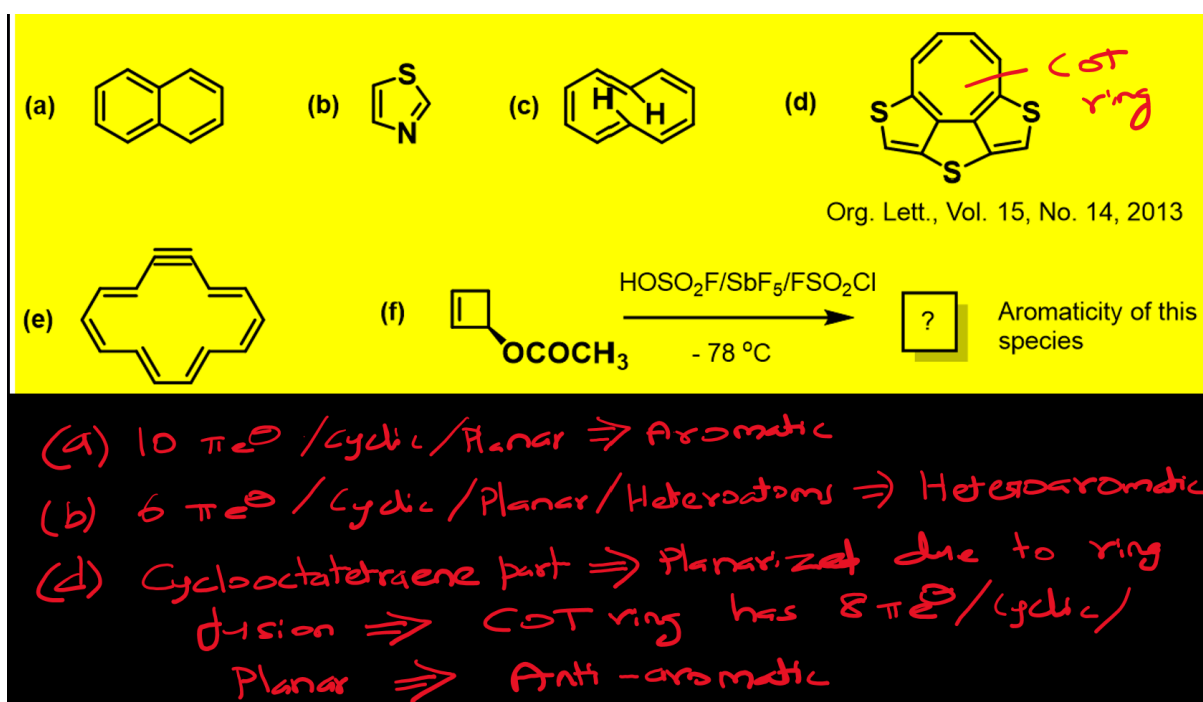
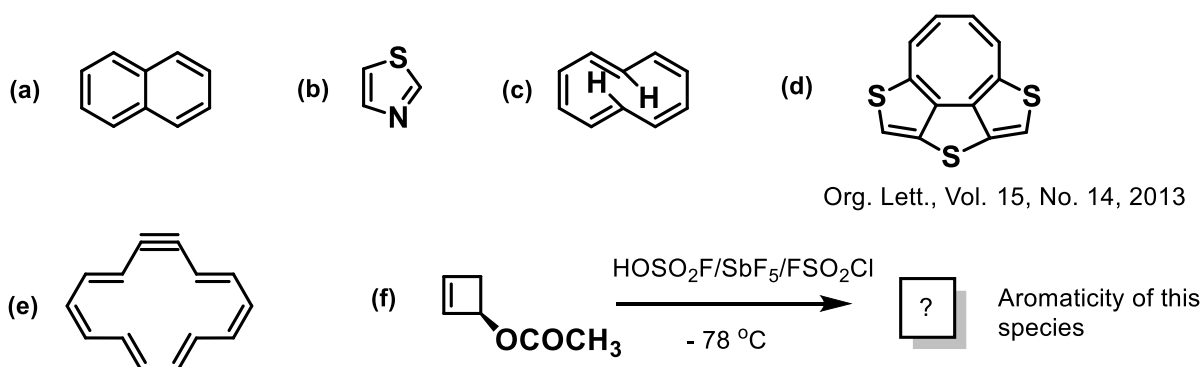
each case and indicate the reason.

(a)	Basicity	
(b)	Ionization of the C-Br bond	
(c)	Lewis acidity	
(d)	Dipole moment	

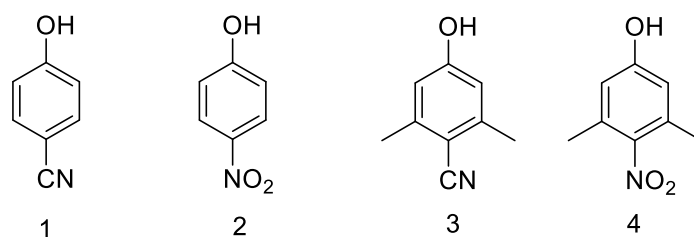
(a) Cyclopentadienyl cation
(b) Cyclopentadienyl radical
(c) Cyclopentadienyl anion

	(a) Cyclopentadienyl cation	(b) Cyclopentadienyl radical	(c) Cyclopentadienyl anion
π_4			
π_5			
π_2			
π_3			
π_1			
(Common MO's for "q & b & c")			

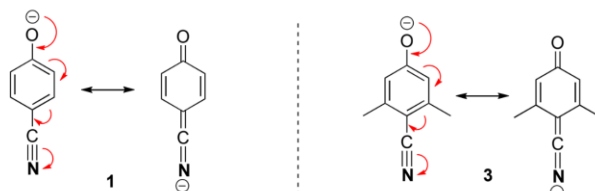
Q13. Classify the following molecules according to their aromaticity characteristics:



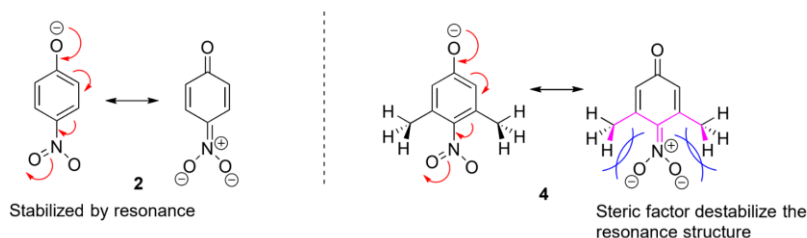
Q14. The pK_a of 1 and 3 are nearly the same, but 2 has lower pK_a value than that of 4. Why?



For lower pK_a , the conjugate base should be stabilized.



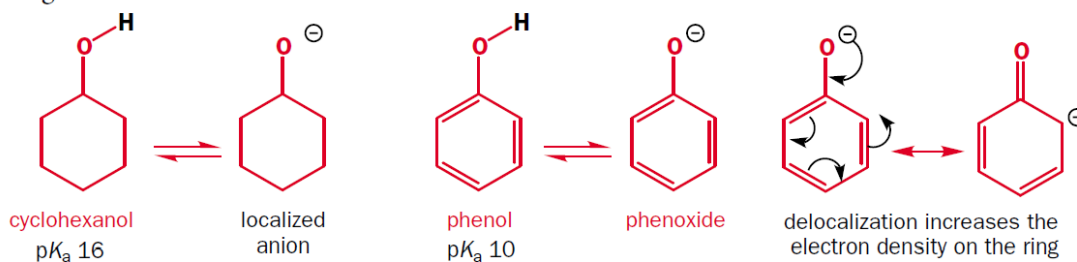
Resonance stabilization due to cyano group is independent of the presence of *ortho* methyl groups in both phenoxide ions. So, their pK_a is nearly equal.



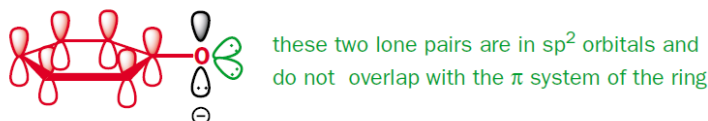
So, **2 (lower pK_a)** is more acidic than **4**.

Q15. Explain why phenol is more acidic ($pK_a = 10$) than cyclohexanol ($pK_a = 16$).

In phenol, $PhOH$, the OH group is directly attached to a benzene ring. On deprotonation, the negative charge can again be delocalized, not on to other oxygens but this time on to the aromatic ring itself.

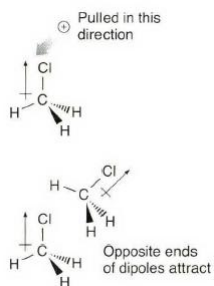


the lone pair in the p orbital can overlap with the π system of the ring, spreading the negative charge on the oxygen on to the benzene ring



The effect of this is to stabilize the phenoxide anion relative to the conjugate base of cyclohexanol where no delocalization is possible and this is reflected in the pK_a s of the two compounds: 10 for phenol but 16 for cyclohexanol.

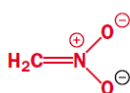
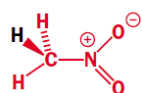
Q16. Predict the possible dimeric interaction mode in CH_3Cl . (Hint: Use the bond dipole(s)/molecular dipole moment for prediction of the charge centres)



Q17. Nitromethane ($\text{pK}_a = 10$) is more acidic than methane ($\text{pK}_a = 48$). Why?



~48 charge is localized on one carbon—again very unsatisfactory



10 charge is delocalized but mainly on oxygens of nitro group